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1

Describe how oxygen in the air reaches capillaries surrounding alveoli in the lungs. Details of breathing are **not** required.

[4 marks]

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[Extra space]

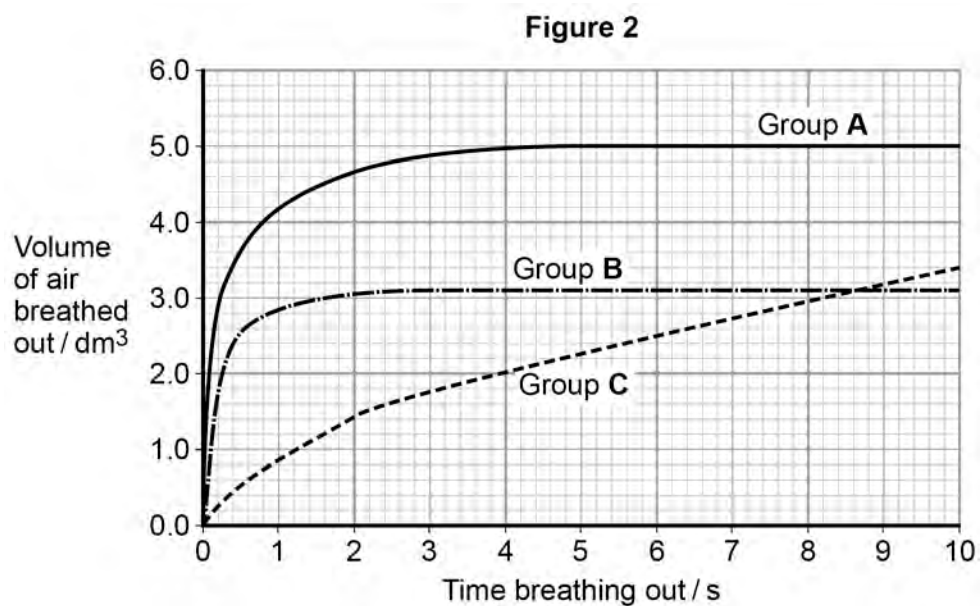
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Forced expiratory volume (FEV) is the greatest volume of air a person can breathe out in 1 second.

Forced vital capacity (FVC) is the greatest volume of air a person can breathe out in a single breath.

**Figure 2** shows results for the volume of air breathed out by three groups of people, **A**, **B** and **C**. Group **A** had healthy lungs. Groups **B** and **C** had different lung conditions that affect breathing.



0 2 . 2

Calculate the percentage drop in FEV for group **C** compared with the healthy people.

[1 mark]

Answer = \_\_\_\_\_

0 2 . 3

Asthma affects bronchioles and reduces flow of air in and out of the lungs.  
Fibrosis does not affect bronchioles; it reduces the volume of the lungs.

Which group, **B** or **C**, was the one containing people with fibrosis of their lungs?  
Use the information provided and evidence from **Figure 2** to explain your answer.

[3 marks]

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[Extra space]

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